MTech major project · MTP2

Computational edition of the Surya Siddhanta

A Python library, traceable learning, and interactive numerical evaluation

Ashutosh Nigam · M25AI2006
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Project objectives and contribution

Implement

Translate the computable procedures across all fourteen chapters into a reusable Python library.

Explain

Connect a chosen date to intermediate values, historical rules and verse references.

Evaluate

Expose the existing DE440 comparison with explicit frame, time and provenance information.

New additions: eight guided learning paths and the full Research Explorer

One calculation core, several interfaces

Source texts → **Chapter functions** → **Structured results**

React + FastAPI	**Chat + tool registry**	**External MCP client**
Calculation and book routes	Same Python functions for numerical answers	Same tool registry through stdio transport

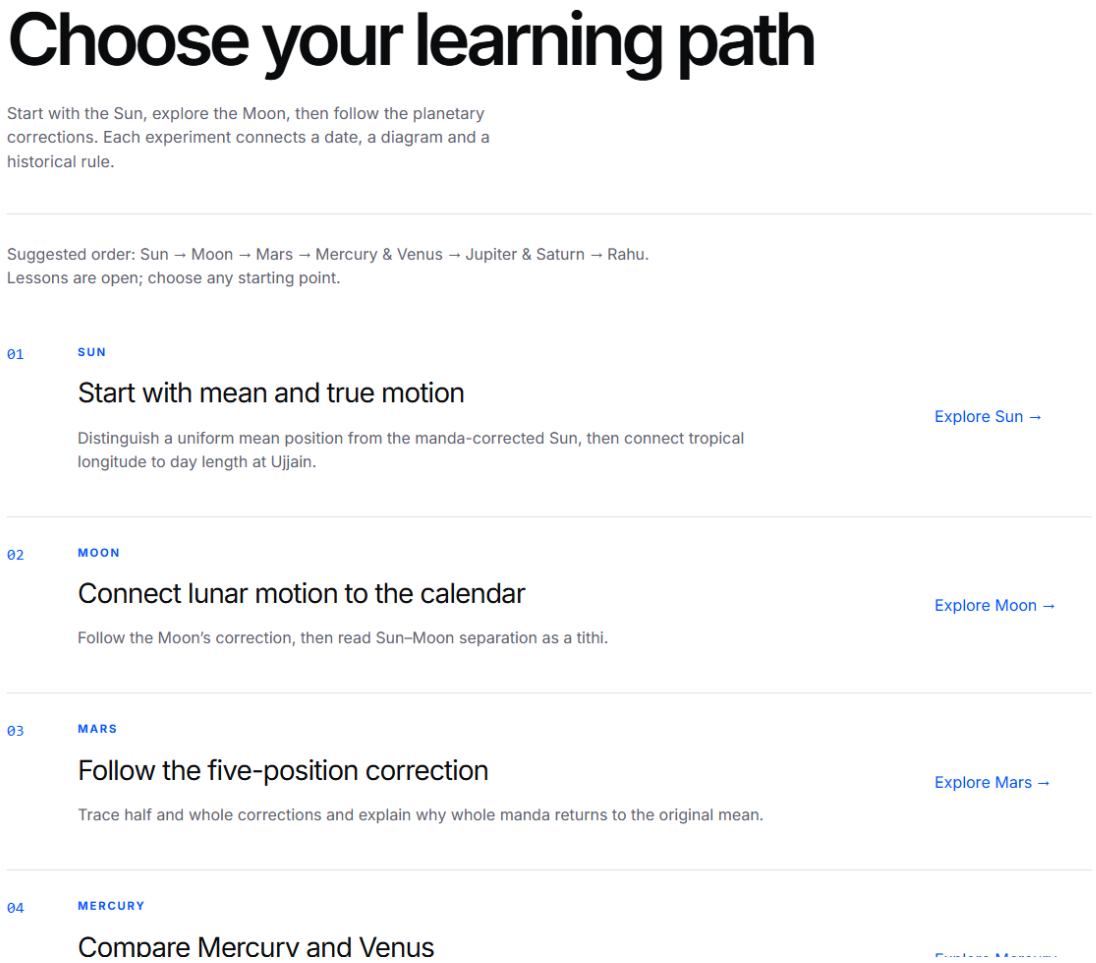
Saved DE440 data feeds the Research Explorer separately from the calculation core.

Computational scope across fourteen chapters

Chapters	Implemented subject areas
1–3	Mean motion, true positions, place and time
4–6	Eclipse calculations and projected geometry
7–11	Conjunctions, stars, heliacal events, Moon and patas
12–14	Cosmography, instruments and reckoning of time

Chapter modules remain the numerical source of truth; interfaces do not reimplement astronomy.

Eight guided learning paths



Sun and Moon

Mean and true motion, seasons, elongation and tithi

Five planets

Correction stages and paired comparisons on the same date

Rahu

Lunar nodes and eclipse geometry

The correction sequence is inspectable

Stage	Sidereal longitude
Mean place	66.9862°
Half sighra	87.1212°
Half manda	91.0859°
Whole manda on the original mean	74.3197°
Whole sighra, final position	114.3004°

4 November 2026 · Ujjain mean time

The fourth step returns to the original mean. “True” denotes the corrected SS-model position.

Research Explorer

OVERVIEW

Where does the error grow?

Time-aligned | Naïve time

Each cell shows RMS longitude error in degrees. Select a cell to inspect its evidence. Colour intensity uses one shared scale across this matrix.

Time-aligned · RMS error (degrees) · darker cells indicate larger errors

Body	1550 −1649	1650 −1749	1750 −1849	1850 −1949	1950 −2049	2050 −2149	2150 −2249	2250 −2349	2350 −2449	2450 −2549	2550 −2649
Sun	1.14	1.27	1.39	1.53	1.66	1.79	1.93	2.51	5.30	8.41	11.54
Moon	1.69	1.76	1.85	1.95	2.05	2.16	2.29	2.79	5.39	8.46	11.58
Mercury	3.31	3.25	3.20	3.16	3.14	3.13	3.14	3.45	5.78	8.71	11.75
Venus	4.16	4.74	5.38	5.96	6.61	7.23	7.77	8.55	10.17	12.58	15.14
Mars	1.88	2.21	2.67	2.96	3.36	3.85	4.16	4.95	7.98	11.20	14.56
Jupiter	2.09	2.24	2.45	2.87	3.33	3.84	4.29	4.19	2.12	1.94	4.45
Saturn	4.88	4.90	5.70	6.35	7.44	8.77	9.45	10.67	13.56	16.65	20.04
Rahu	1.50	1.79	2.10	2.42	2.76	3.10	3.43	3.43	1.63	2.16	4.56

Rahu is the ascending lunar node. The last windows include future extrapolations. These are sampled longitude errors, not eclipse timing errors.

Time conventions change the comparison

Moon · 1550–1649 CE

Treatment	RMS error	Mean signed error
Naïve time	3.990°	-3.732°
Time-aligned	1.686°	-0.975°

The reference epoch changes; the historical SS equations stay the same.

The time-aligned variant includes the Ujjain meridian conversion and an approximate Delta-T model.

Long-term error differs by body

Body	1950–2049	2250–2349	2550–2649
Sun	1.658°	2.506°	11.536°
Moon	2.045°	2.791°	11.580°
Mars	3.356°	4.953°	14.559°
Venus	6.608°	8.555°	15.142°
Saturn	7.436°	10.671°	20.043°

RMS of tropical longitude residuals · future dates are model extrapolations

Verification and reproducibility

1,653

Python tests passed

176

body / window / treatment groups checked against saved tables

Browser checks

Eight lessons, date changes, retry, source links, exports and mobile layout

Portable evidence

JSON and CSV include source hashes, conventions and sample counts

Limits of the evidence

Historical interpretation	The adopted implementation covers computable procedures, with documented choices and remaining gaps.
Numerical comparison	Ten-day sampling can miss extrema between samples. Longitude error does not validate event timing.
Educational use	The understanding check is implemented. Its effect on student learning has not been measured.

Live demonstration

1	**Choose a learning path**	Explore Sun, Moon and the correction stages.
2	**Explain the geometry**	Compare tithi, paired planets or lunar nodes.
3	**Compare the model**	Select a matrix cell and switch time treatments.
4	**Export the evidence**	Download reproducible statistics and conventions.

localhost:5173/learn · localhost:5173/research

Contribution and next work

An executable and inspectable computational edition

The new interfaces connect the historical rule, its intermediate calculation and the evidence used to evaluate it.

Next: measure learning outcomes and continue the source-based algorithm audits.

Sources and implementation references are recorded in the speaker notes.